

S-DES : DES SIMPLIFICADO

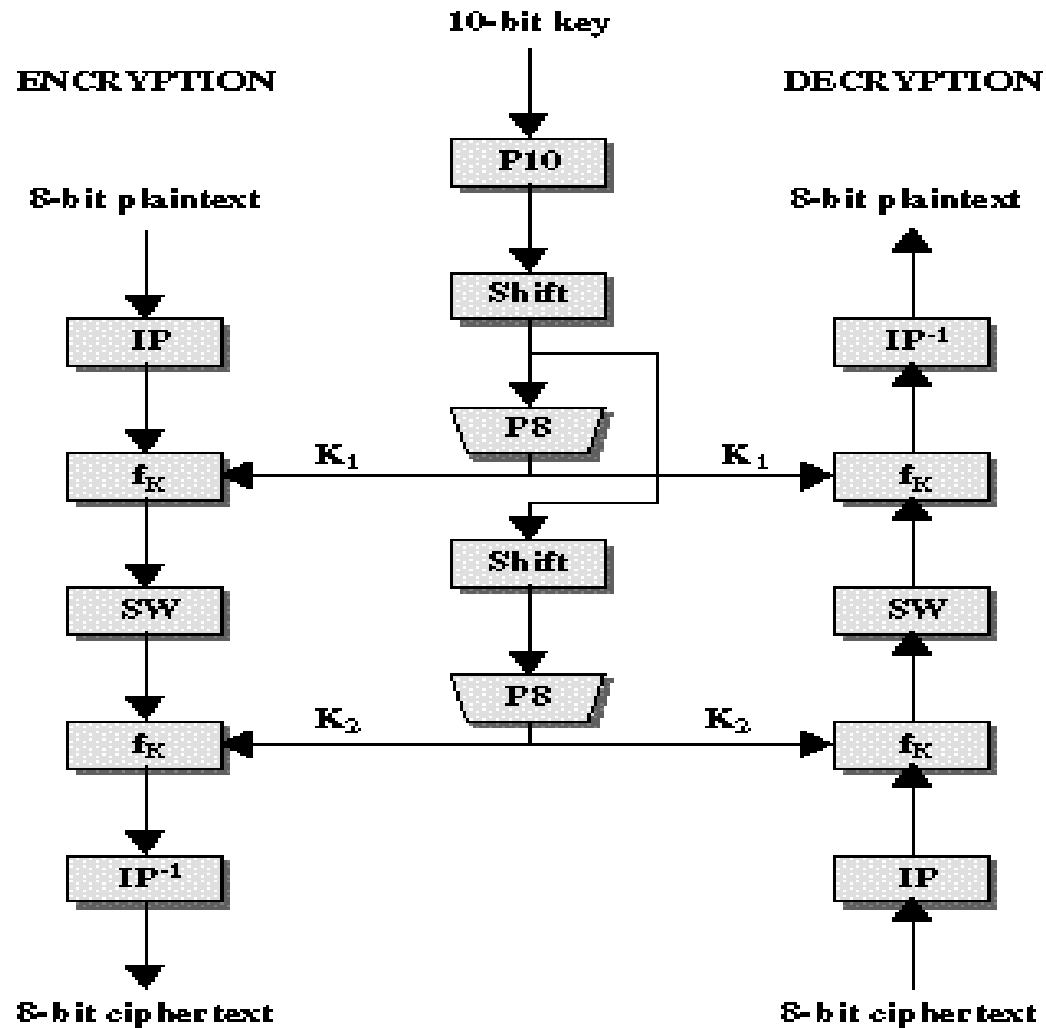
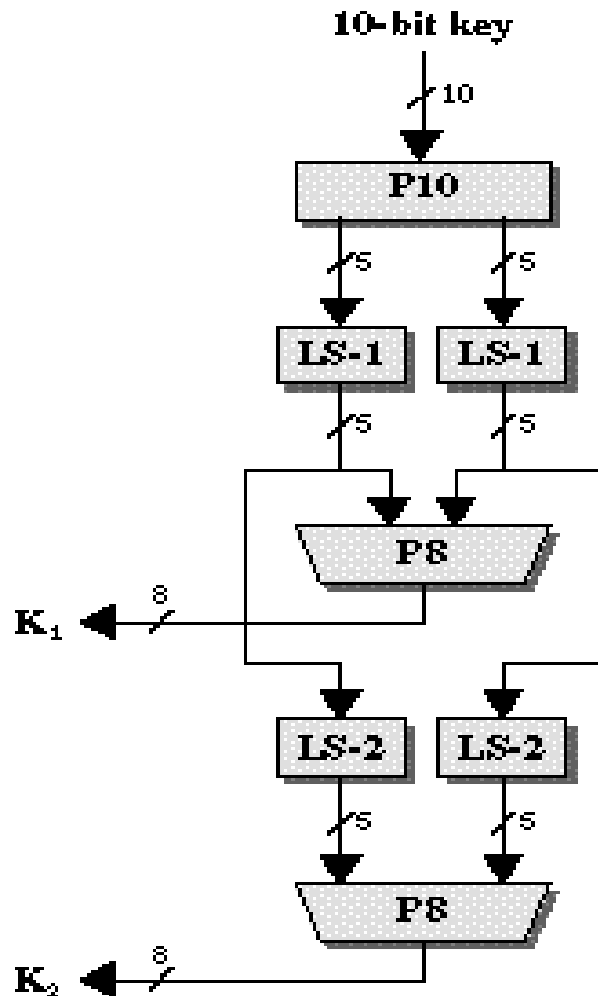


Figure 3.1 Simplified DES Scheme

S-DES: GERAÇÃO DAS SUB-CHAVES K1 e K2



$P_{10} = \{3, 5, 2, 7, 4, 10, 1, 9, 8, 6\}$

$P_8 = \{6, 3, 7, 4, 8, 5, 10, 9\}$

LS-1 = Deslocamento circular à esquerda de uma posição.

LS-2 = Deslocamento circular à esquerda de duas posições.

Exemplo:

$K = 10100\ 00010$

$P_{10} = 10000\ 01100$

LS-1 $00001\ 11000 \Rightarrow P_8 \Rightarrow K_1 = 1010\ 0100$

LS-2 $00100\ 00011 \Rightarrow P_8 \Rightarrow K_2 = 0100\ 0011$

Figure 3.2 Key Generation for Simplified DES

PROCESSO DE ENCRIPÇÃO

$f_K(L,R) = L \text{ XOR } F(R,K_i)$, $R \rightarrow$ Encriptar/Decryptar

$IP = \{ 2, 6, 3, 1, 4, 8, 5, 7 \}$ e $IP^{-1} = \{ 4, 1, 3, 5, 7, 2, 8, 6 \}$

$E/P = \{ 4, 1, 2, 3, 2, 3, 4, 1 \}$

$P4 = \{ 2, 4, 3, 1 \}$

$S0 = \begin{matrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{matrix}$

$S1 = \begin{matrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{matrix}$

$n1n2n3n4 \rightarrow S_i[n1n4][n2n3]$

Exemplo:

$R = 1010$ $E/P = 0101 \ 0101$

$K1 = 1010 \ 0100$

$XOR = 1111 \ 0001$

$S0[11][11] \rightarrow 10$ $S1[01][00] \rightarrow 10 \ (1010)$

$P4 = 0011$

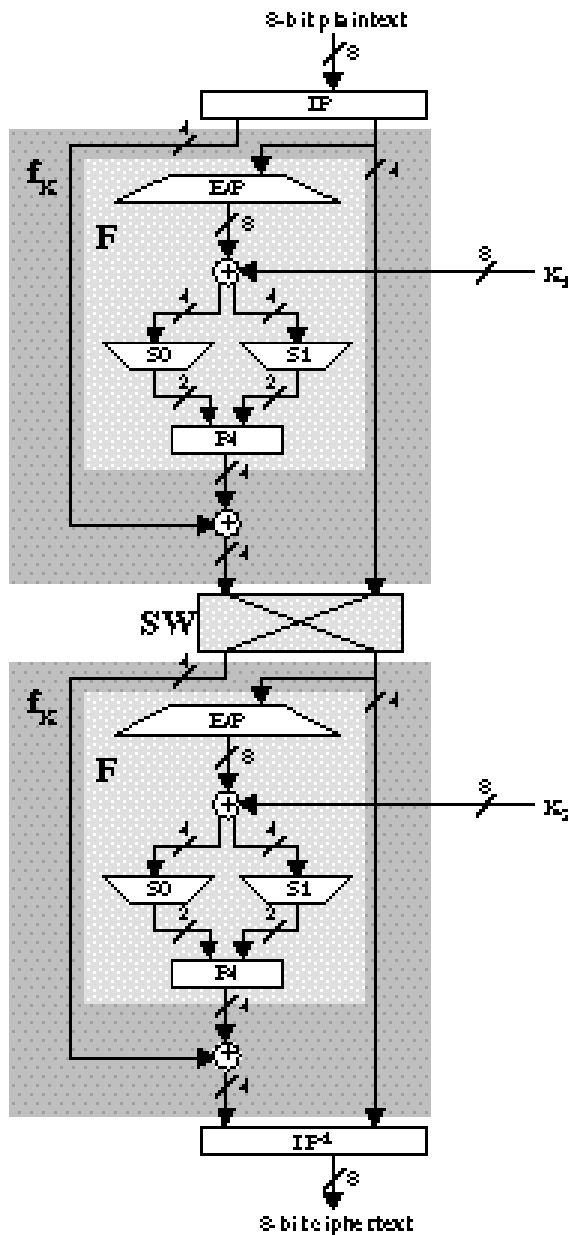


Figure 3.3 Simplified DES Encryption Detail